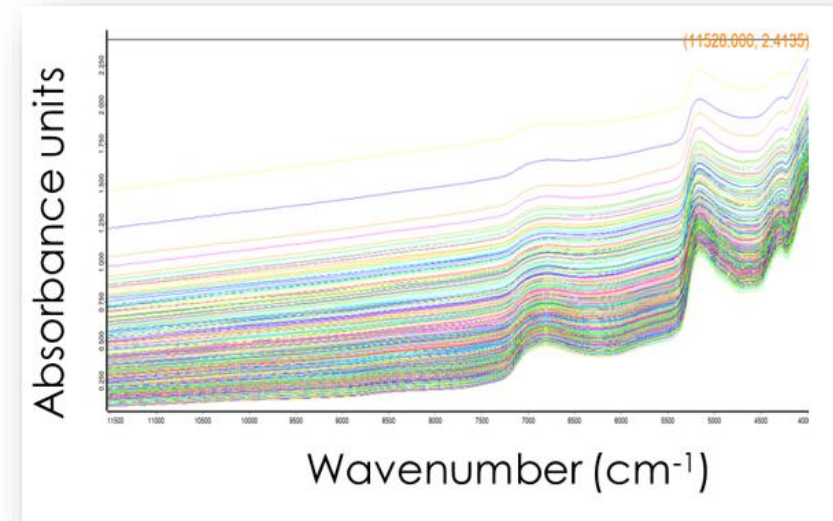
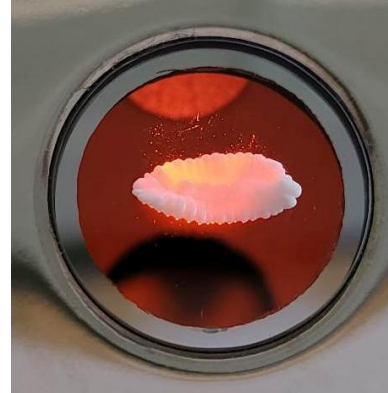


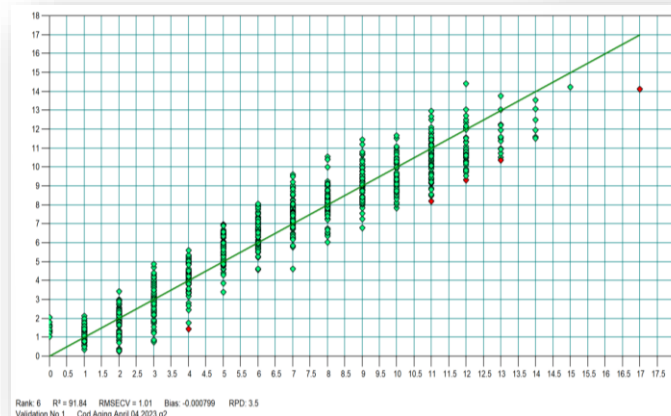
Investigating the use of FT-NIR spectroscopy for aging marine mammals and fish and for species identification



Aaron Adamack^{1*}, Kelly Antaya¹, Sudagar Dhaliwal², Karen Dwyer¹, Shelley Lang¹, Christina Bourne¹, Kim Emond¹

¹ Fisheries and Oceans Canada

² Bruker



Acknowledgements

- Countless scientific staff, harvesters, and Coast Guard crew who were involved in the collection of the otoliths and seal teeth
- Many technicians across several sections for all their work in aging Atlantic cod, Atlantic herring, and Harp seals and collecting samples for this study
- Kelly Antaya who did the bulk of the work in developing our scanning protocol and in scanning the otoliths and teeth used for this study
- Bruker for the loan of a FT-NIR spectrometer for this study
- This project was funded by a grant from the Competitive Science Research Fund at DFO
- My coauthors for all of their help in making this project happen.

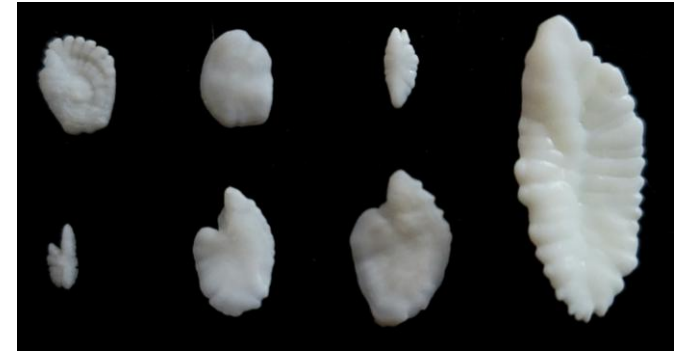
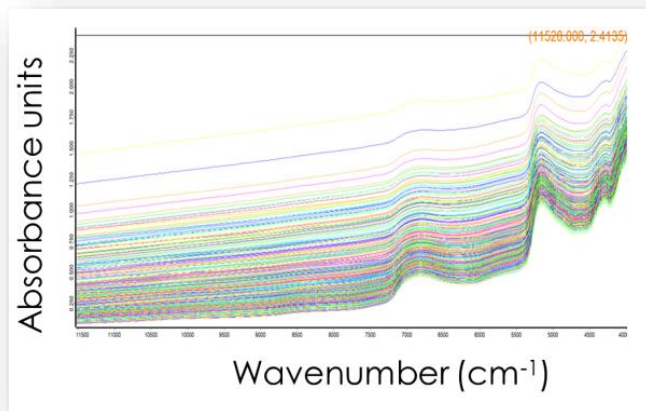
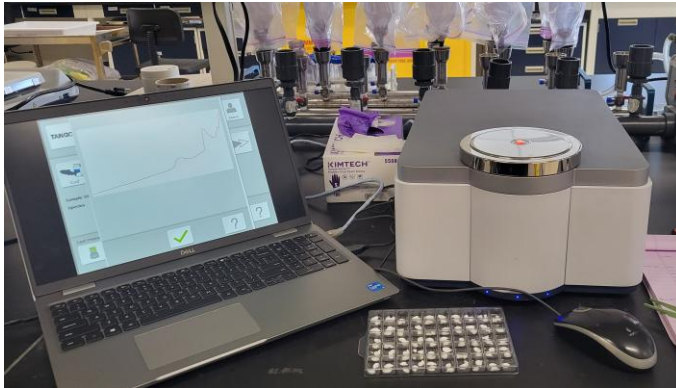


Photo by K. Antaya

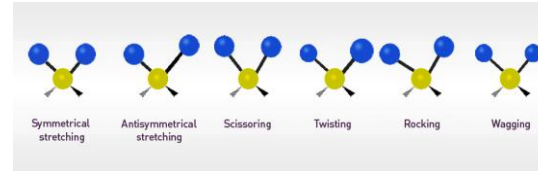
Aging fish using FT-NIR

- FT-NIR spectrometry is new for aging otoliths
 - started in 2010s; ~20 - 50 published papers
- Recent work by Australian and American researchers has demonstrated the potential of FT-NIR for aging otolith
 - Robins et al. 2015
 - Helser et al. 2019; Matta and Helser 2024
- Used in other fields for decades
 - Chemistry, food science, agriculture, health sciences, manufacturing, etc.
- The overall method is very well established with an extensive literature on statistical methods, and practices

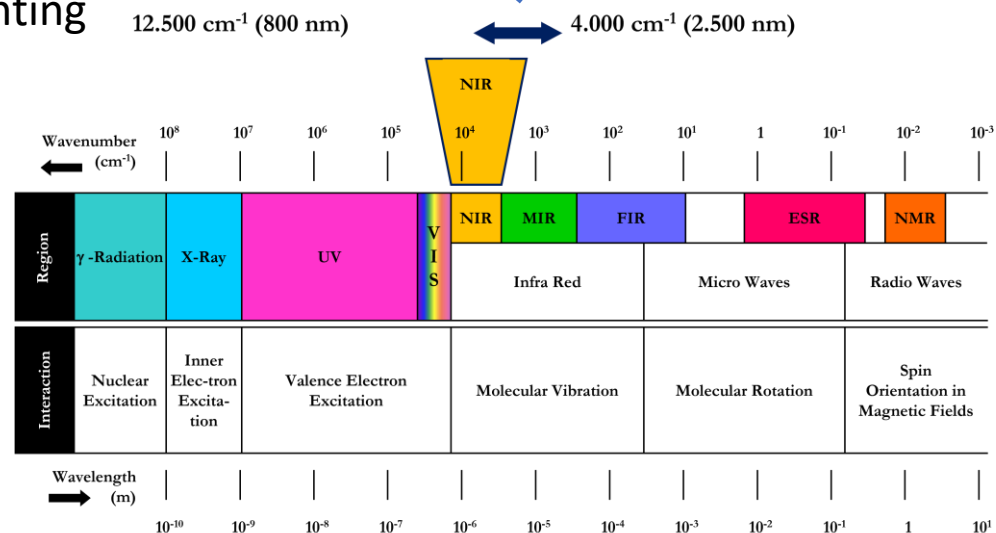
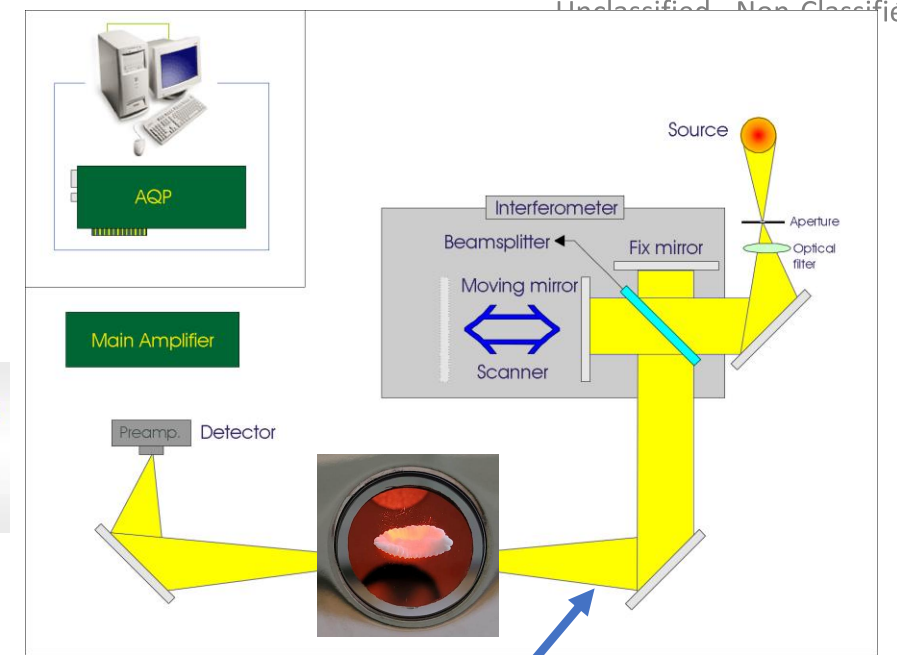
How FT-NIR works



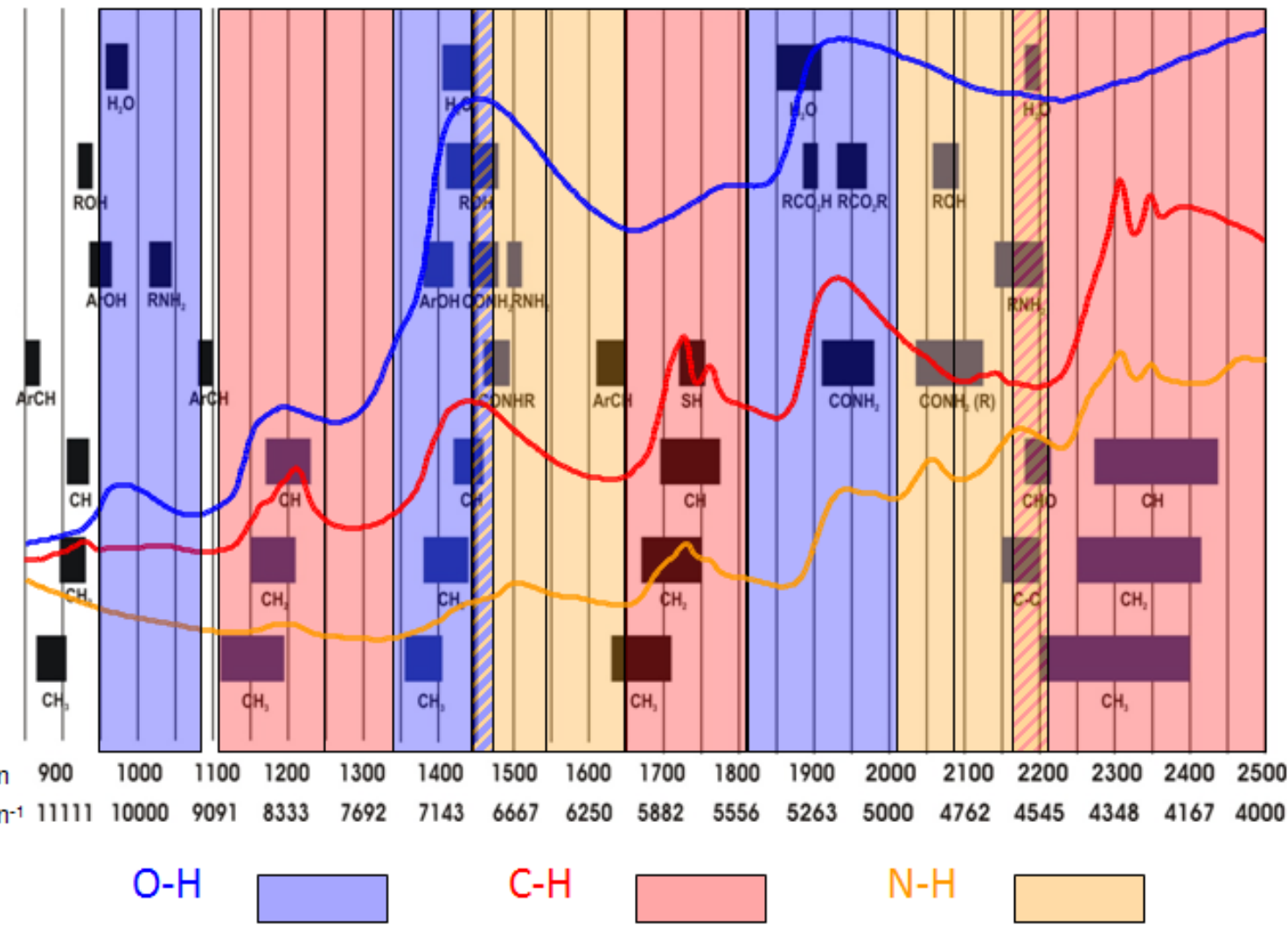
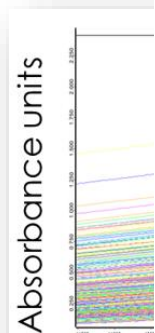
Diagrams from Bruker
Pictures and figures from
Adamack and Antaya



Spectrometer detects
otolith mass, chemical
composition (CaCO_3 and
protein) and physical
structure. NOT counting
annuli

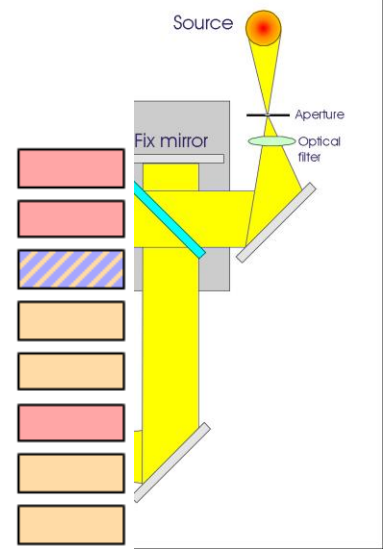


Passeroti et al. 2022



for C-H and N-H

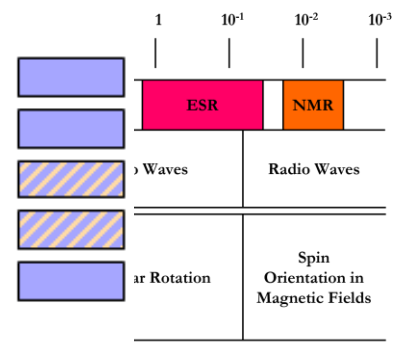
9000	-	8000	cm ⁻¹
8000	-	7450	cm ⁻¹
6900	-	6770	cm ⁻¹
6770	-	6400	cm ⁻¹
6400	-	6030	cm ⁻¹
6030	-	5500	cm ⁻¹
4950	-	4770	cm ⁻¹
4770	-	4600	cm ⁻¹
4600	-	4500	cm ⁻¹
4500	-	3850	cm ⁻¹



for O-H

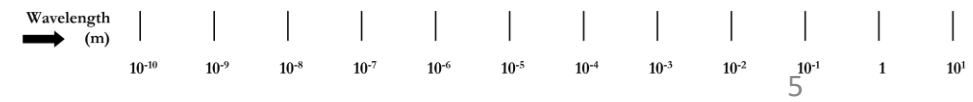
10550	-	9250	cm ⁻¹
7100	-	6800	cm ⁻¹
6800	-	6400	cm ⁻¹
6400	-	6030	cm ⁻¹
5300	-	4950	cm ⁻¹

0 nm)



Diagrams from Duke
Pictures and figures from
Adamack and Antaya

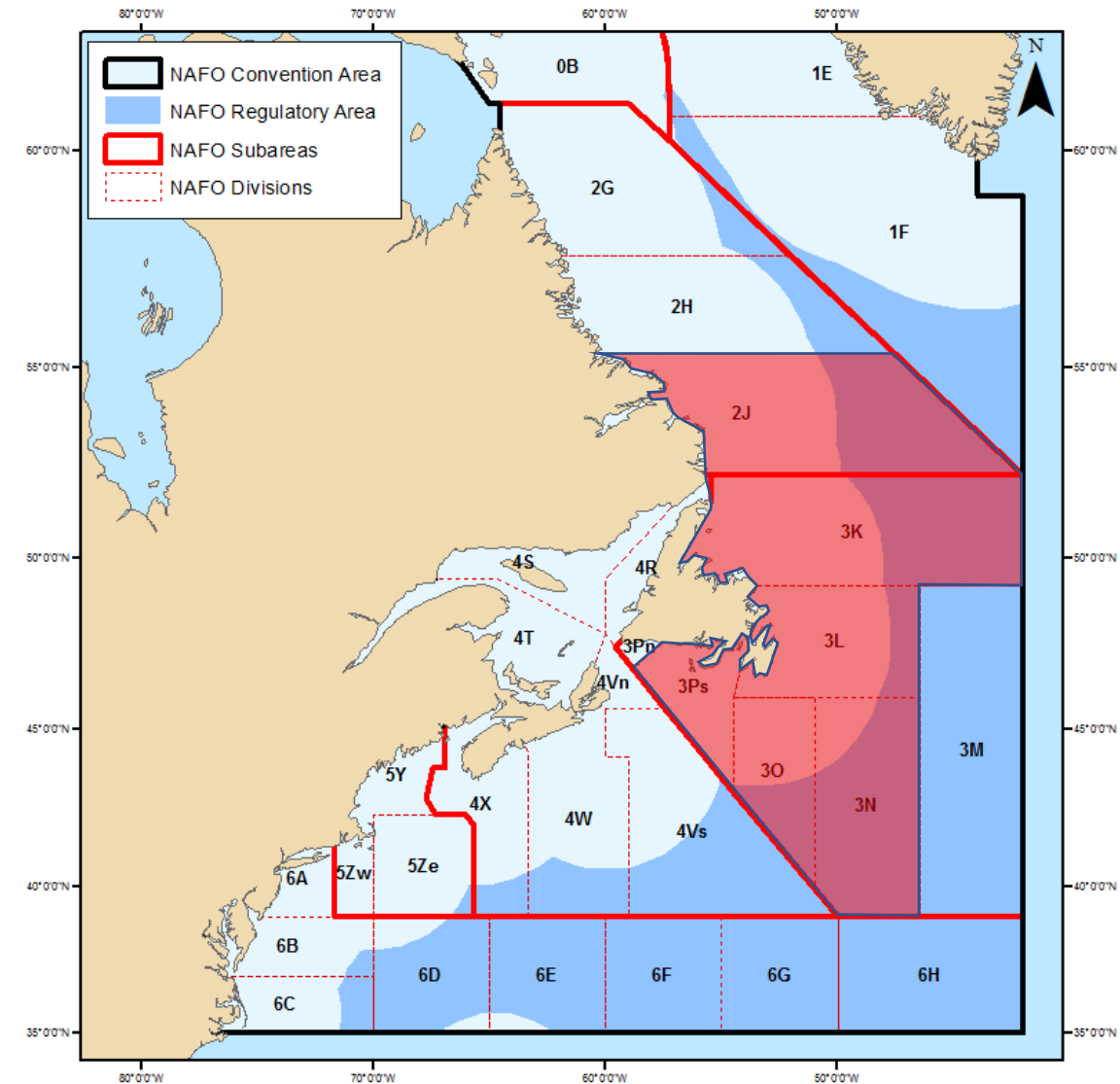
Passeroti et al. 2022



Applying FT-NIR to aging Atlantic cod otoliths

Study objectives

1. Investigate the potential for using FT-NIR to age Atlantic cod otoliths from North Atlantic Fisheries Organization (NAFO) Divisions 2J, 3K, 3L, 3N, 3O, and 3Ps
2. Develop an overall calibration model for all Atlantic cod sampled and separate calibration models for each NAFO Div.
3. Conduct a time-cost analysis investigating some of the economics of adopting this approach

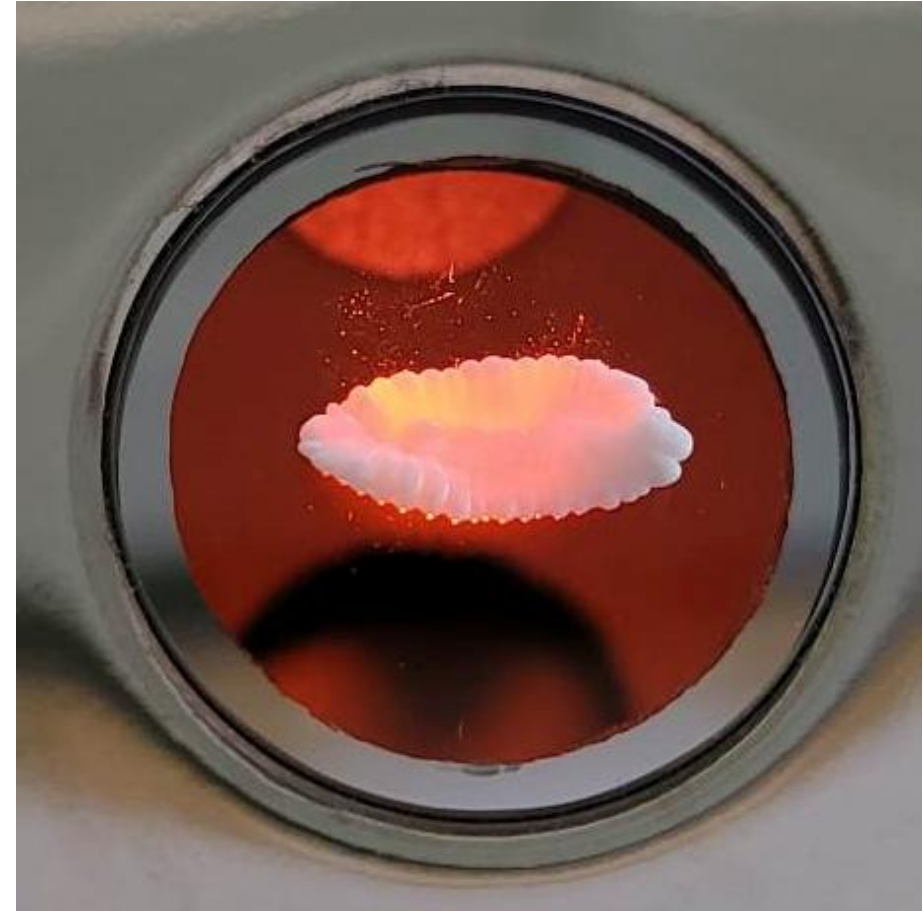


Overview of sampling

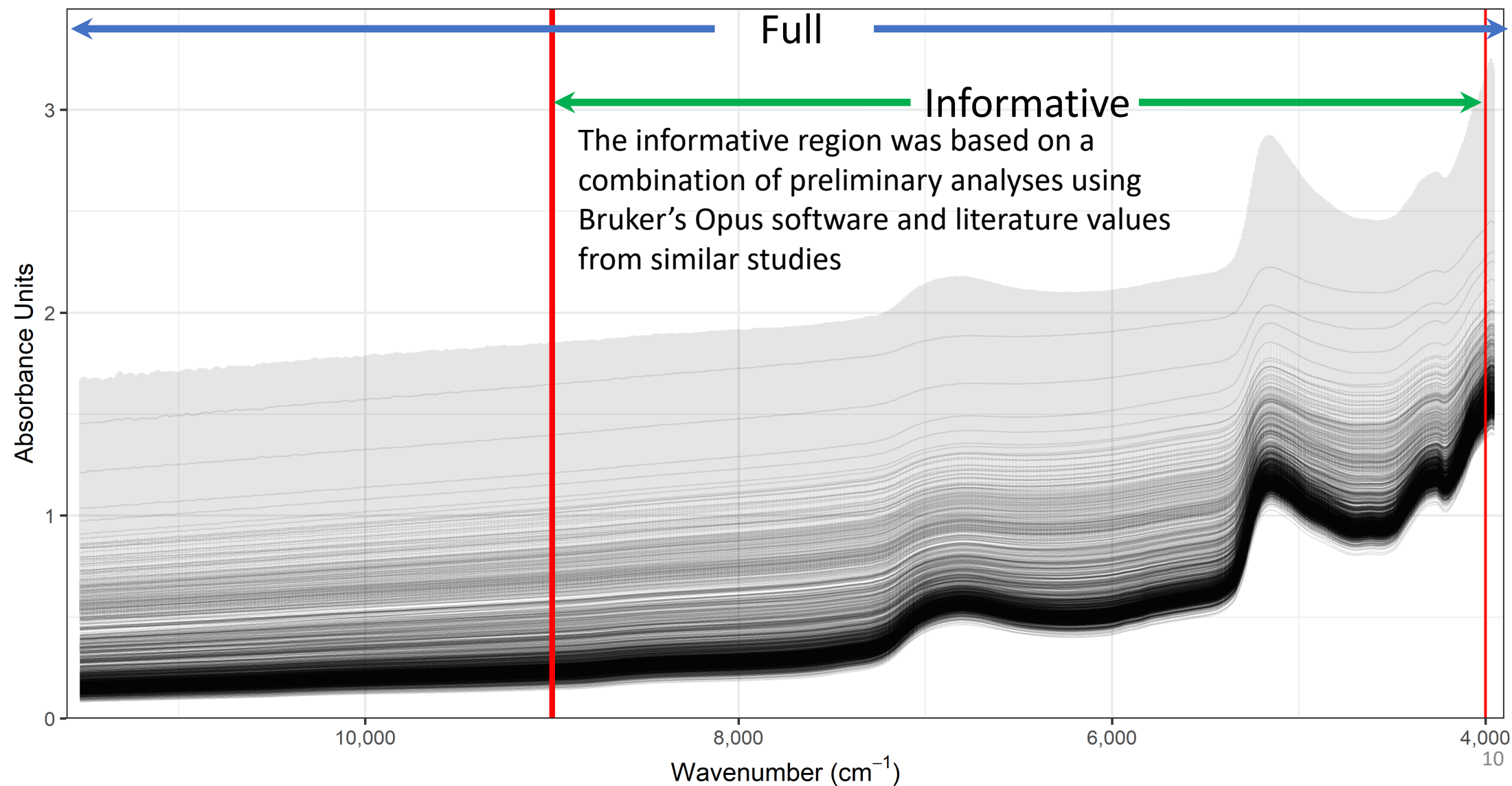
- Used previously collected and aged (by counting growth rings) Atlantic cod otoliths from NAFO Divs. 2J, 3K, 3L, 3N, 3O, and 3Ps for ages 0-12+
- For NAFO Division $\times$ Age $\times$ Sex, attempted to get 6 otoliths
- Close to plan for ages 2-11
- Otoliths for ages 0, 1, and 12+ were underrepresented
- Did not find otoliths for all age classes/sexes for Div. 3O

Overview of sampling

- 805 cod otoliths scanned
- Scanned concave-side up; rostrum facing to the right
- Some otoliths scanned multiple times for analyses not covered here
- For this analysis, I'm using the first scan for each individual unless QA/QC issues arose



Initial data / QA-QC



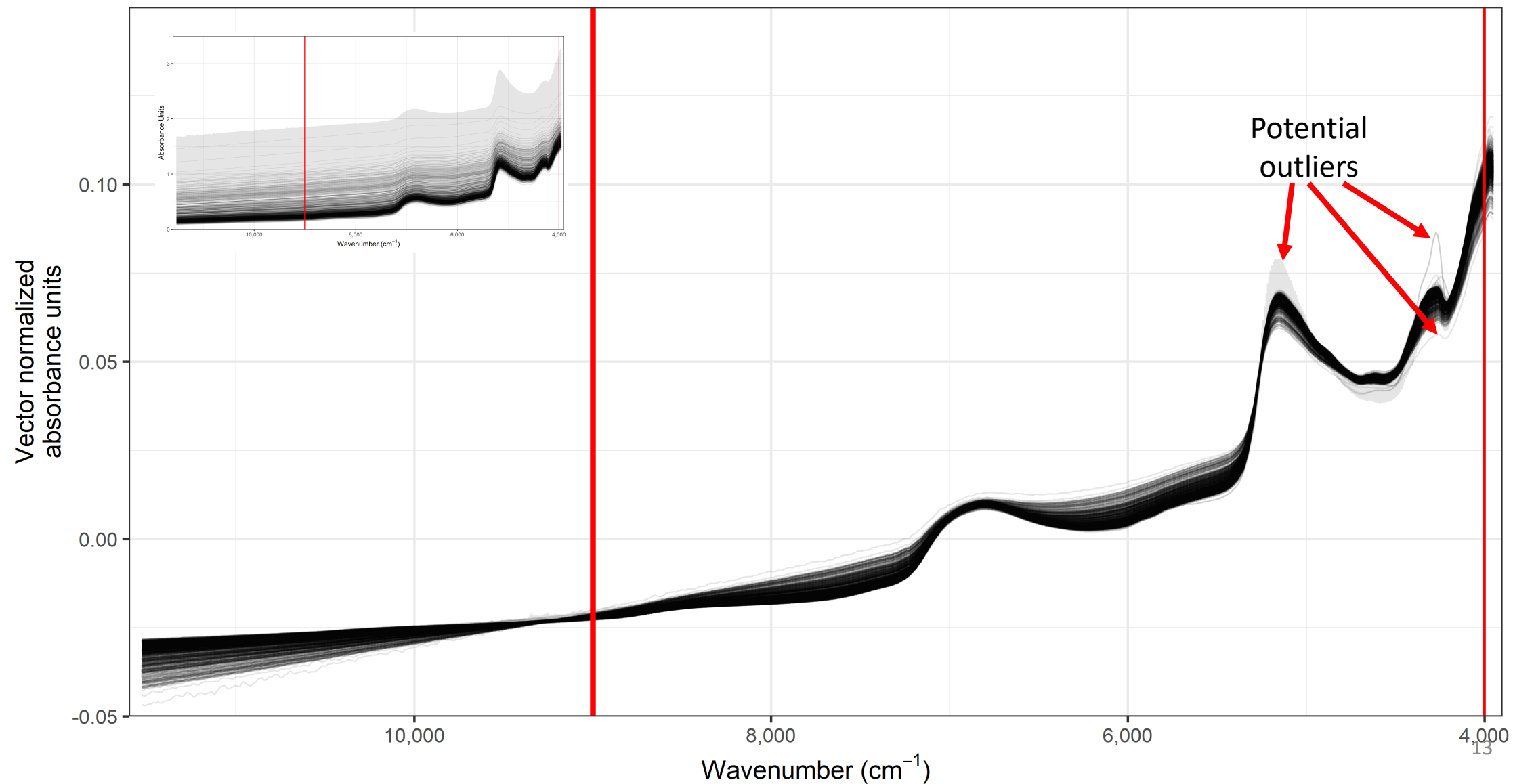


Initial data / QA-QC

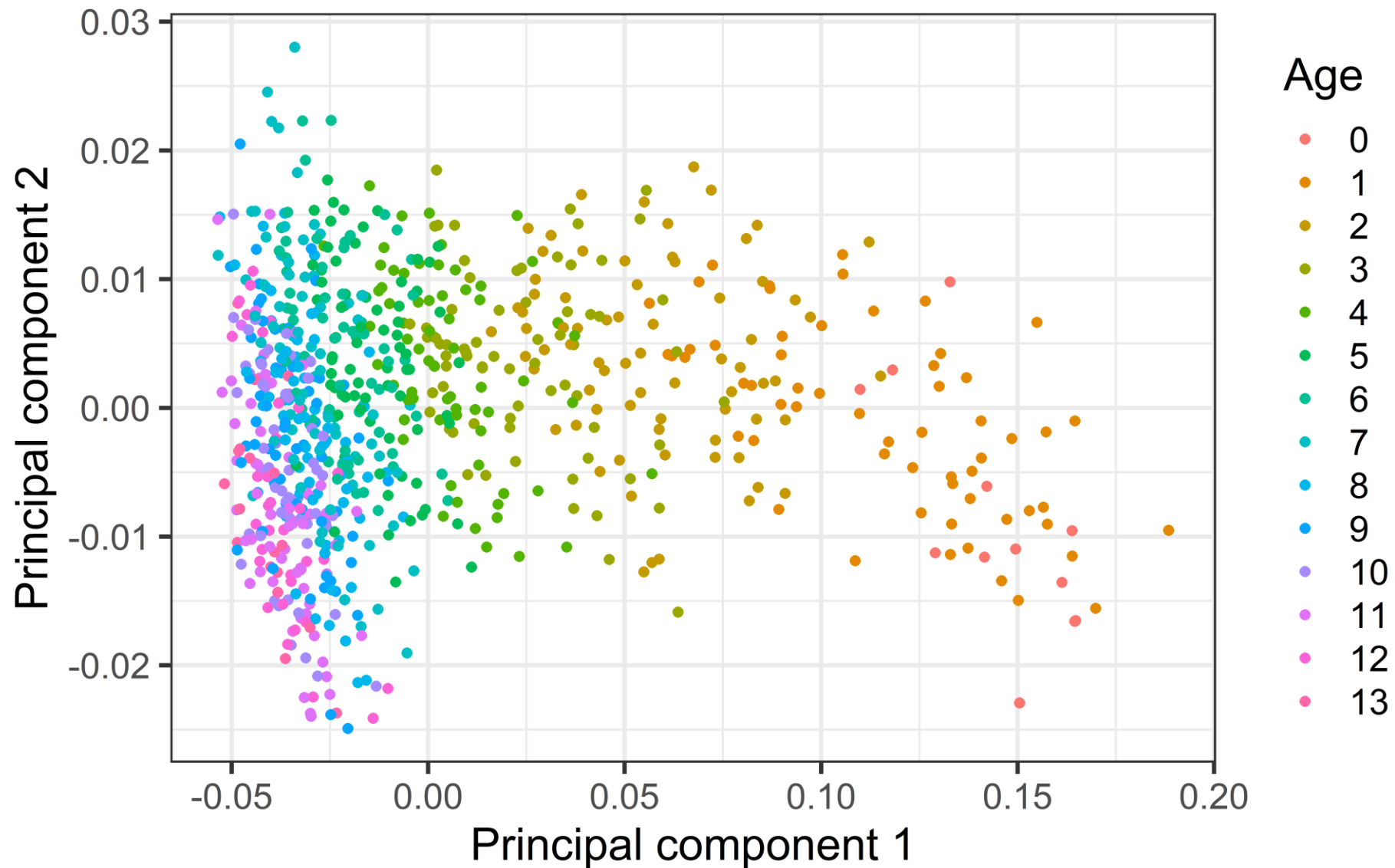
Prior to analysis some Data Quality Assurance/Quality control steps need to be taken:

1. Shifting baseline (e.g. vertical shift in the position of the individual spectra) adjusted for using vector normalization
2. Outliers were detected using PCA analysis.
 - Too difficult to spot potential outliers on spectral signatures
 - Examined plots of PC 1 vs. PCs 2-10 and visually identified outliers
 - 6 spectra from 4 otoliths were identified as outliers
 - 3 individuals dropped from analysis
 - An alternative spectra was available for the 4th otolith

Initial data / QA-QC



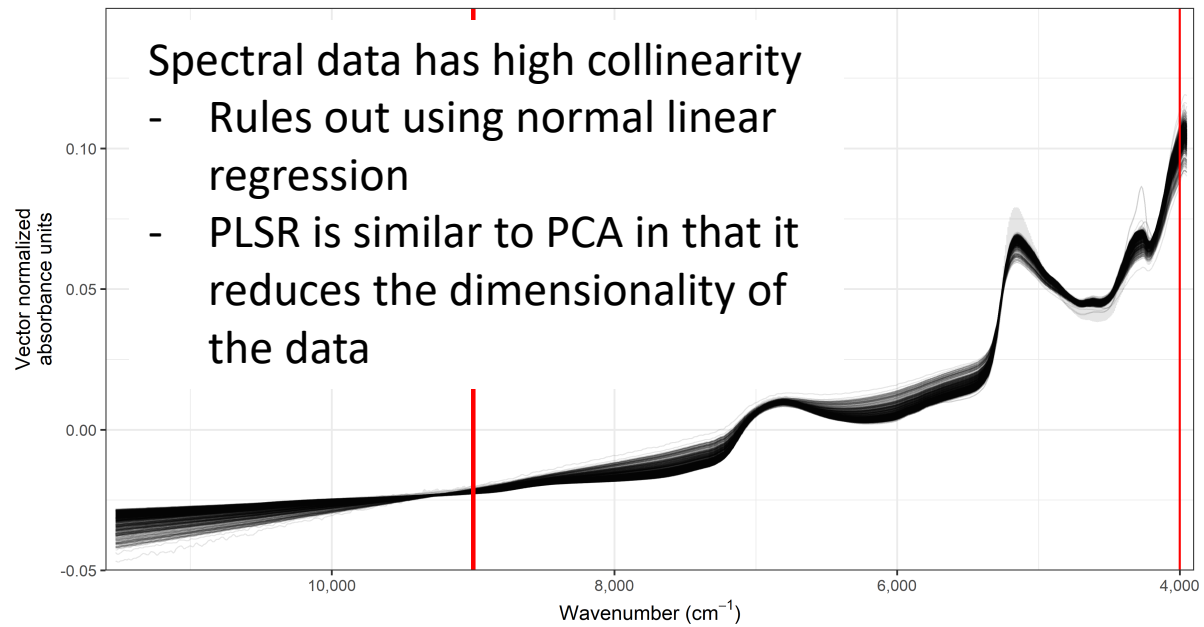
PCA of Atlantic cod FT-NIR spectra



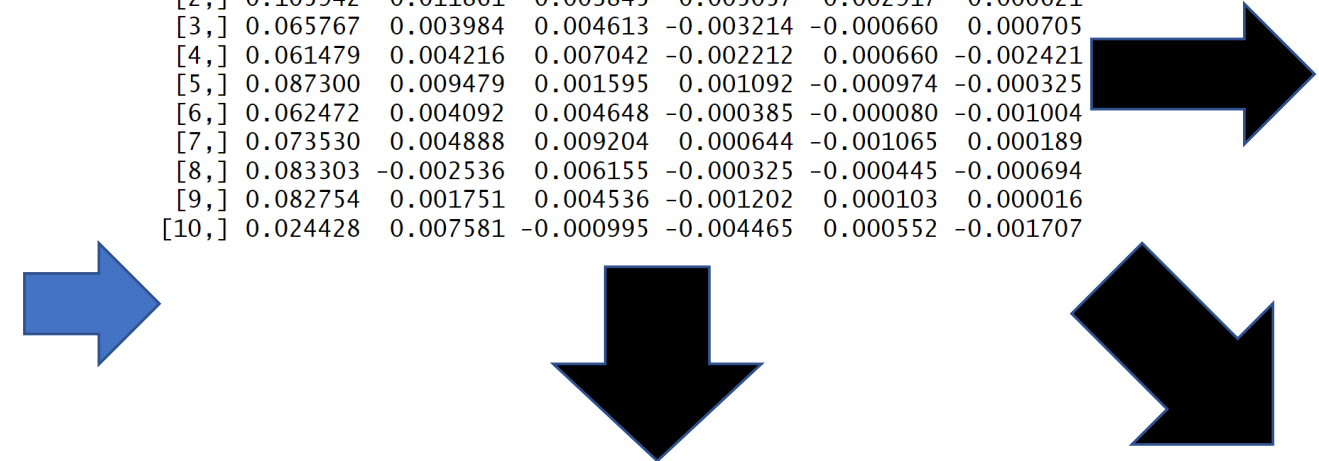
Calibration models

- Atlantic cod were divided into fit (70%) and test (30%) data sets
- Partial least-squares regression models were fit between the spectral signatures for each otolith and known ages from traditional aging using the fit data set with separate models for the full and informative portions of the spectral signatures
 - Overall (all NAFO Divisions)
 - Regional (individual NAFO Divisions)
- Model fit was evaluated by fitting linear regressions between the fitted age and the predicted ages for both the fit and the test data sets
- Examined adjusted R^2 and RMSE across multiple components.

A brief note on PLSR



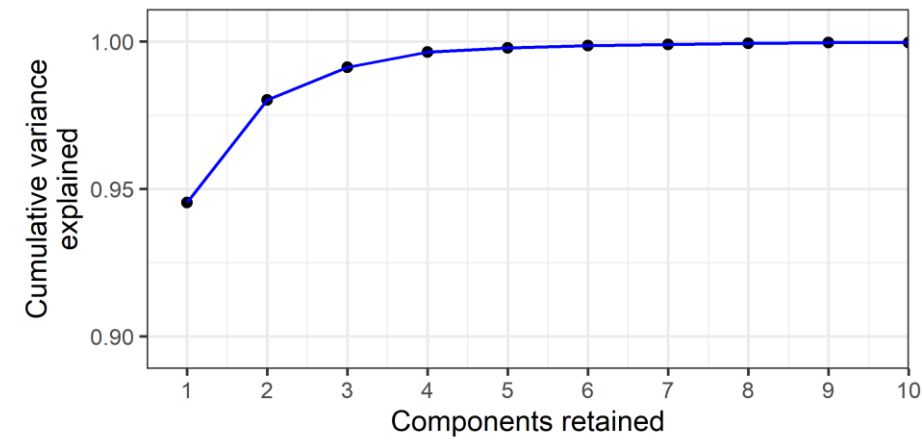
	PC1	PC2	PC3	PC4	PC5	PC6
[1,]	0.069377	0.009833	0.001559	-0.001056	-0.000812	-0.000045
[2,]	0.105942	0.011861	0.003845	0.005057	-0.002917	0.000621
[3,]	0.065767	0.003984	0.004613	-0.003214	-0.000660	0.000705
[4,]	0.061479	0.004216	0.007042	-0.002212	0.000660	-0.002421
[5,]	0.087300	0.009479	0.001595	0.001092	-0.000974	-0.000325
[6,]	0.062472	0.004092	0.004648	-0.000385	-0.000080	-0.001004
[7,]	0.073530	0.004888	0.009204	0.000644	-0.001065	0.000189
[8,]	0.083303	-0.002536	0.006155	-0.000325	-0.000445	-0.000694
[9,]	0.082754	0.001751	0.004536	-0.001202	0.000103	0.000016
[10,]	0.024428	0.007581	-0.000995	-0.004465	0.000552	-0.001707



PLS models try to find the multidimensional direction in X space that explains the maximum multidimensional variance direction in the Y space

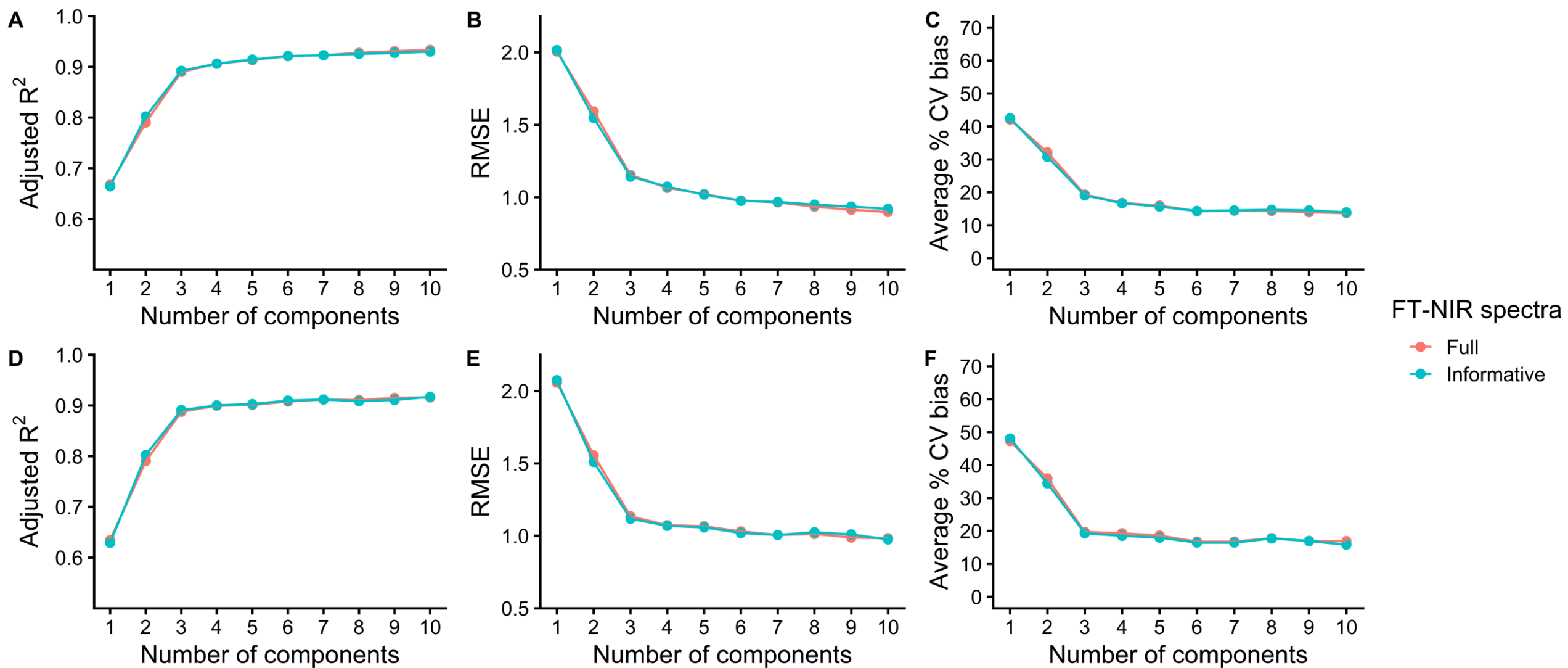
Maximum # of components = number of observations

We could retain all components to get an exact match, but fewer components may be good enough

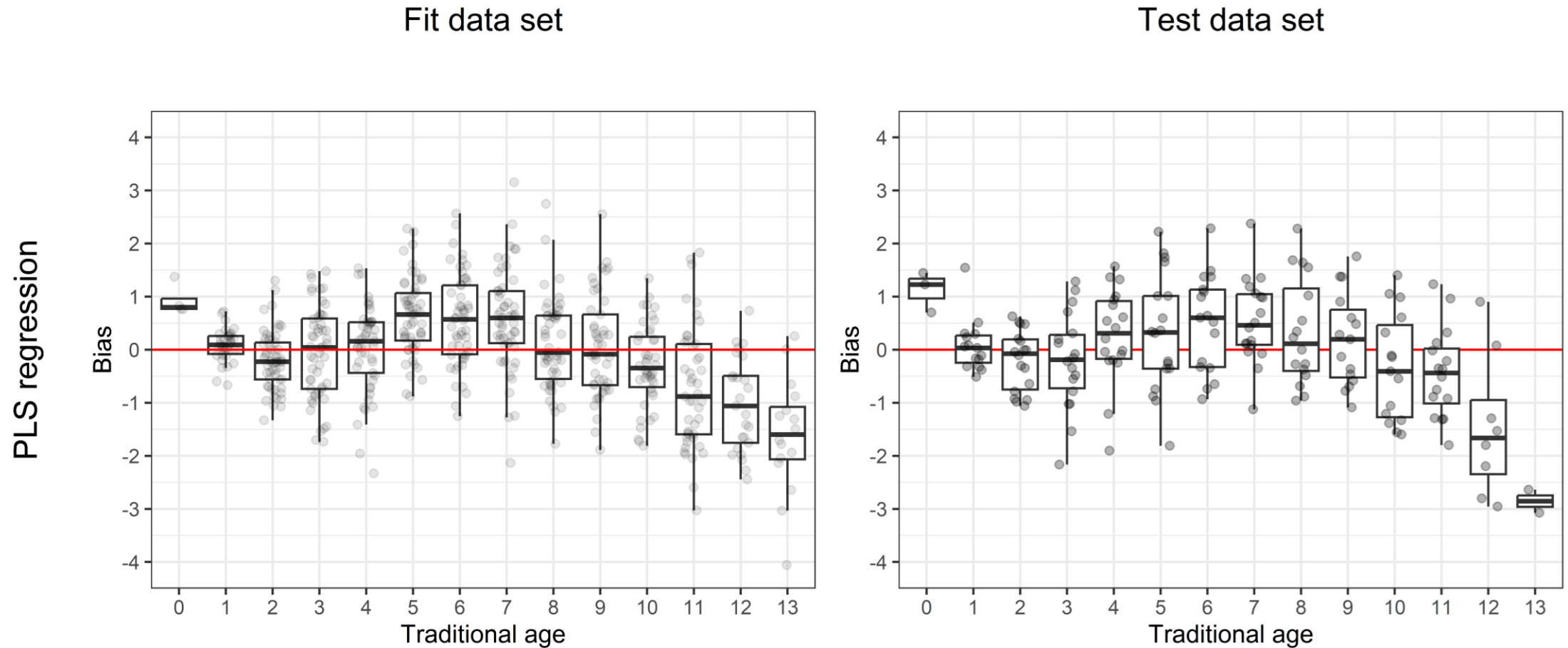


Need to balance between fit and generality to avoid overfitting.

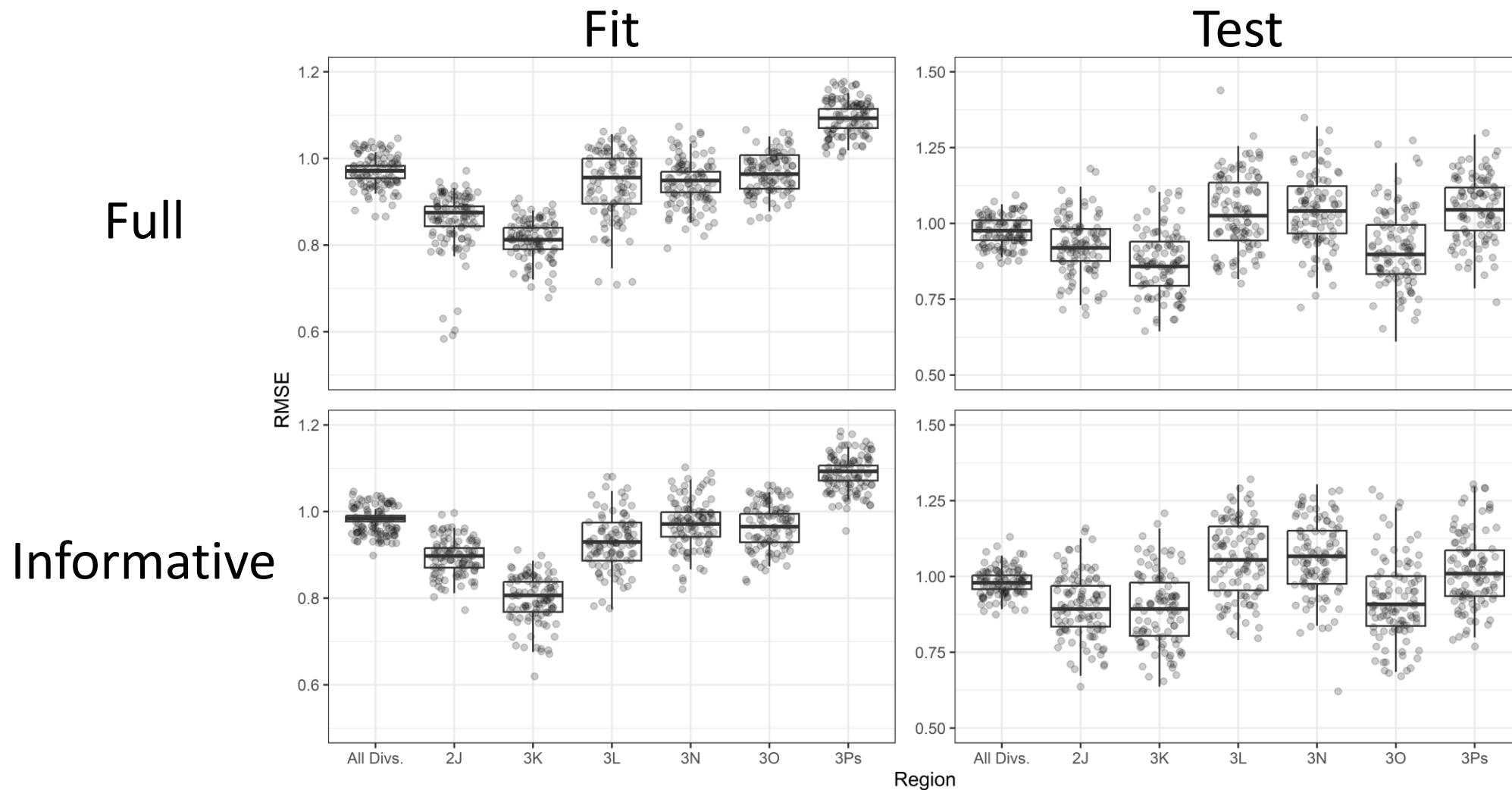
Results – Across NAFO Divisions - PLSR



Results – Across NAFO Divisions - Bias



Results – Individual NAFO Divisions



Time cost analysis

	Existing method	FT-NIRS – Combined NAFO	FT-NIRS – Individual NAFO
Otoliths aged counting annuli	12,000	200	1200
Otoliths aged compared to existing method (%)	N/A	1.67	10
Otoliths scanned using FT-NIRS	N/A	12,000	12,000
Time required to age otoliths counting annuli (days)	160	5.33	32
Scanning time (days)	0	48	48
Cleaning time (days)	0	43	43
Total time required (days)	160	96.33	123
Time cost savings (%)	N/A	40	23

Summary

- We were able to develop good calibration models for Atlantic cod across NAFO Divisions
 - Adjusted $R^2 = 0.92$
 - RMSE = 0.98
 - ACV = 14.3
 - Using only the “informative” region of spectral signatures seems to provide only limited benefit and in some cases may produce a worse fit
- Results for individual NAFO Divisions less clear
 - Insufficient sample sizes to provide definitive results
 - Some NAFO Divs may have better results, others worse
- Aging using FT-NIR is potentially much faster than traditional aging
 - 23 – 40% time cost savings switching from primarily traditional aging to FT-NIRS
- (Open access) Study available at : <https://doi.org/10.1016/j.fishres.2025.107331>

Preliminary analysis using FT-NIR to age harp seal teeth

Harp seal objectives

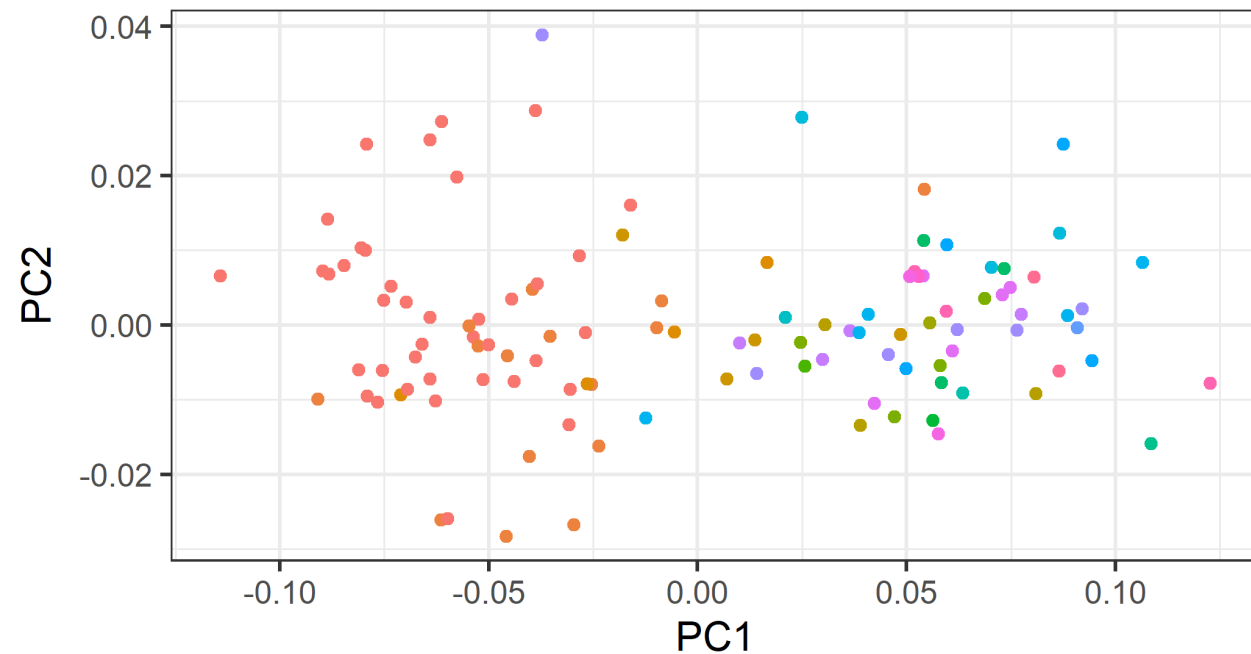
- Proof of concept study
- To my knowledge, there hasn't been work on applying this method to age marine mammal teeth
- Investigating whether canine or premolar teeth work better

Methods

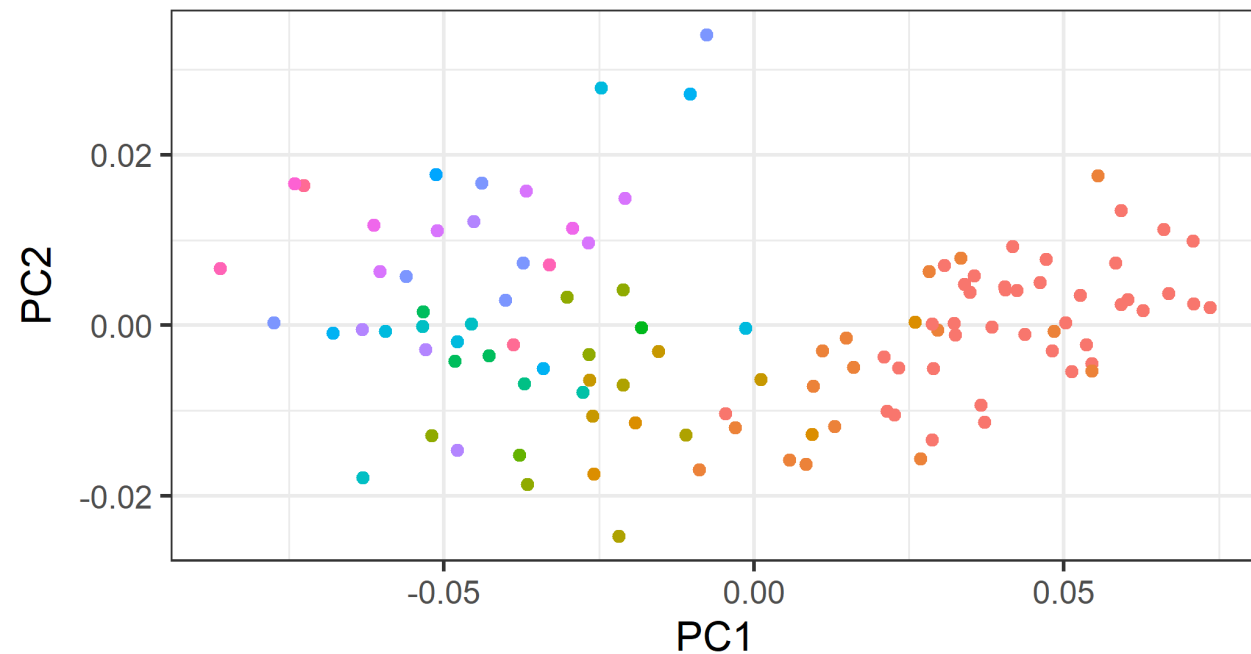
- Methods for harp seal generally followed the same pattern as Atlantic cod
- 114 canine teeth scanned
- 112 premolar teeth scanned
- Ran PCAs for data cleaning
- Fit initial models - PLSR

PCAs

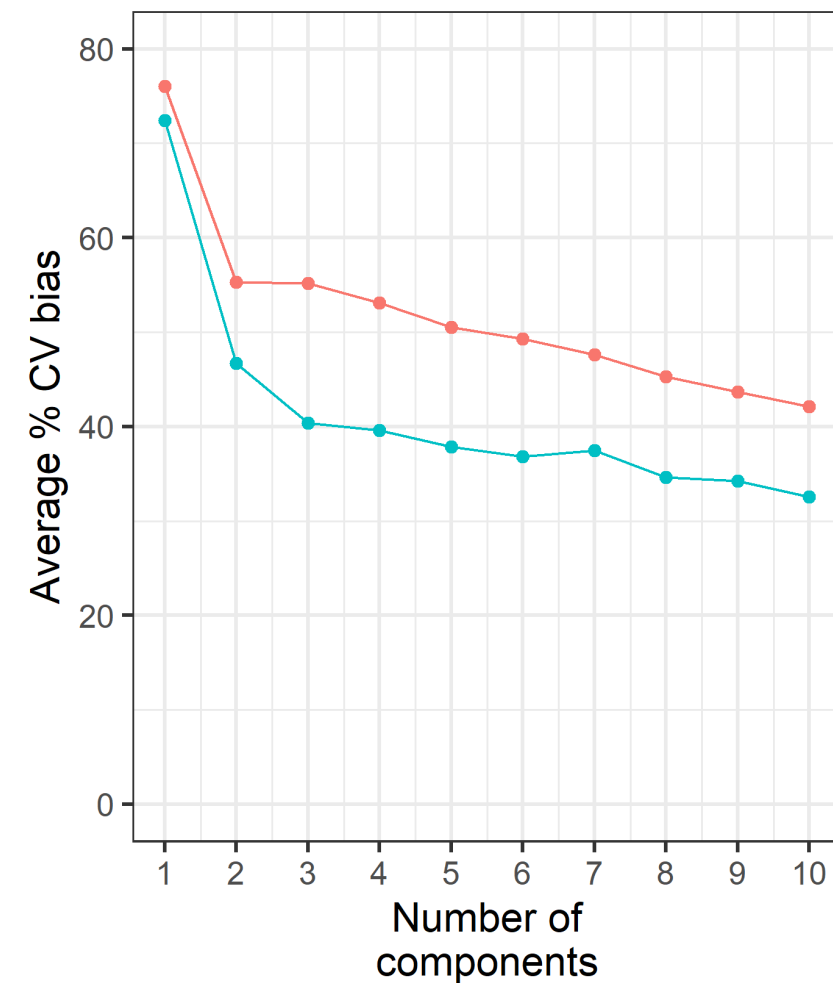
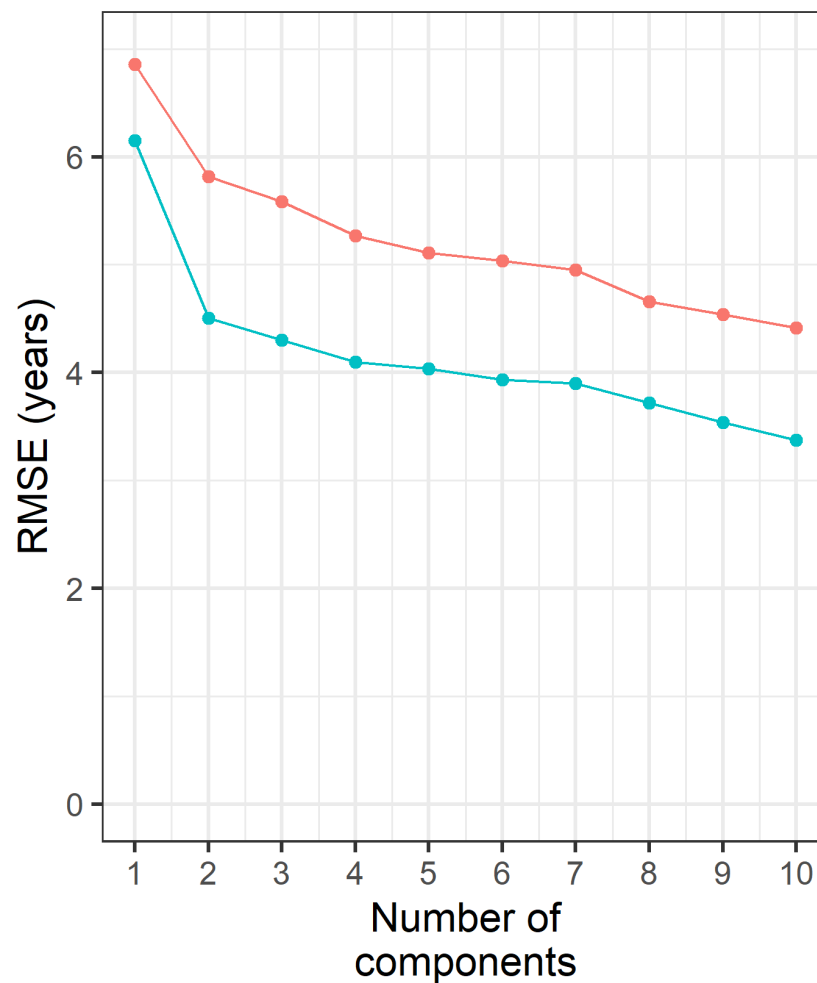
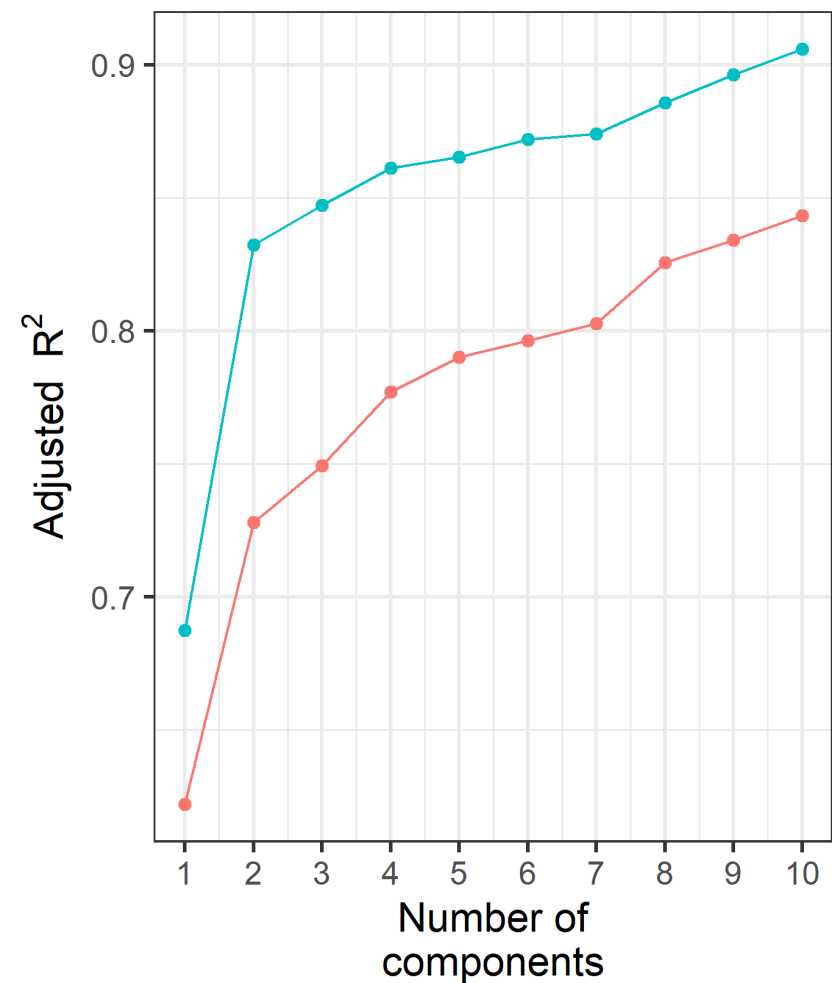
Canine PCA



Premolar PCA



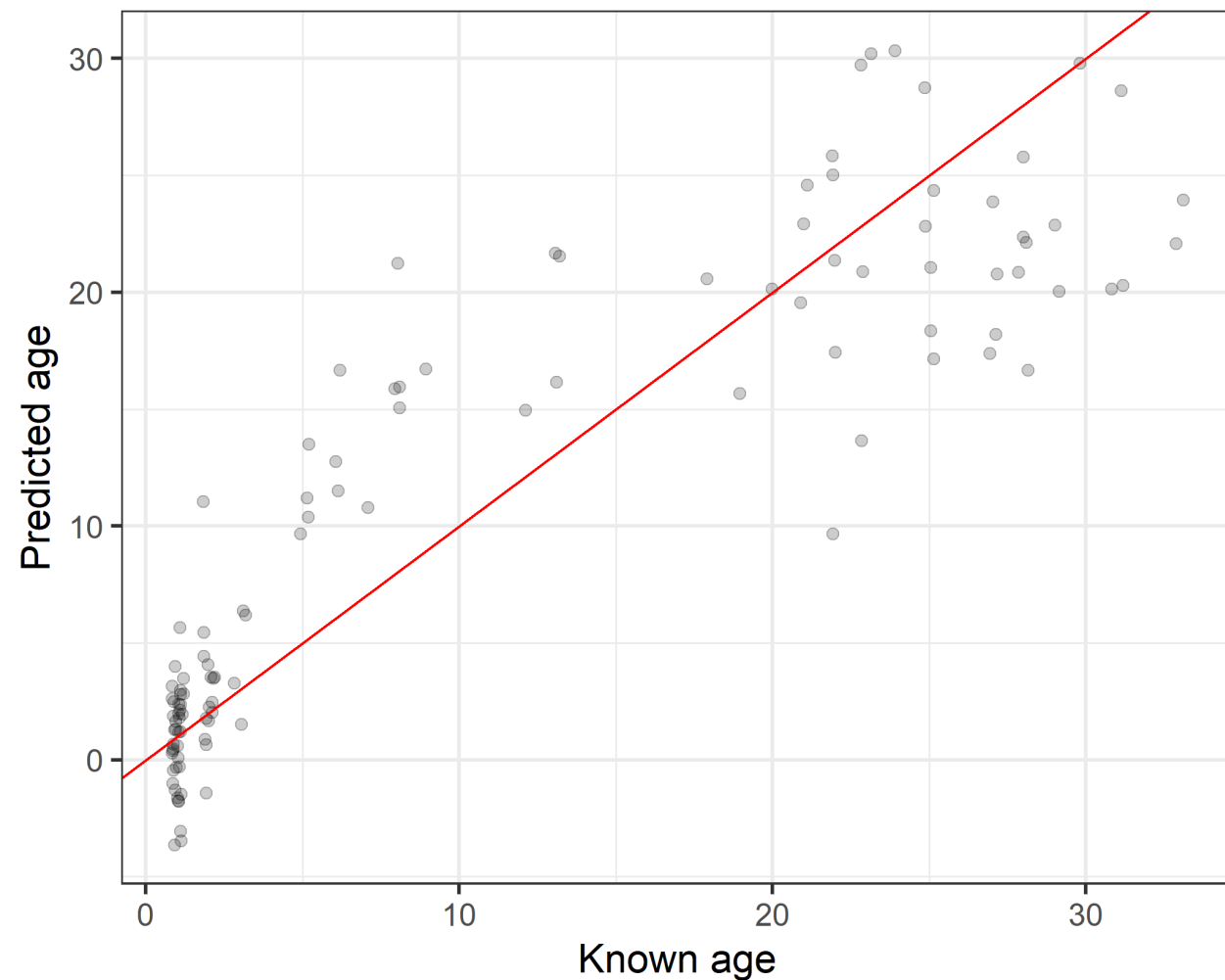
Preliminary seal results



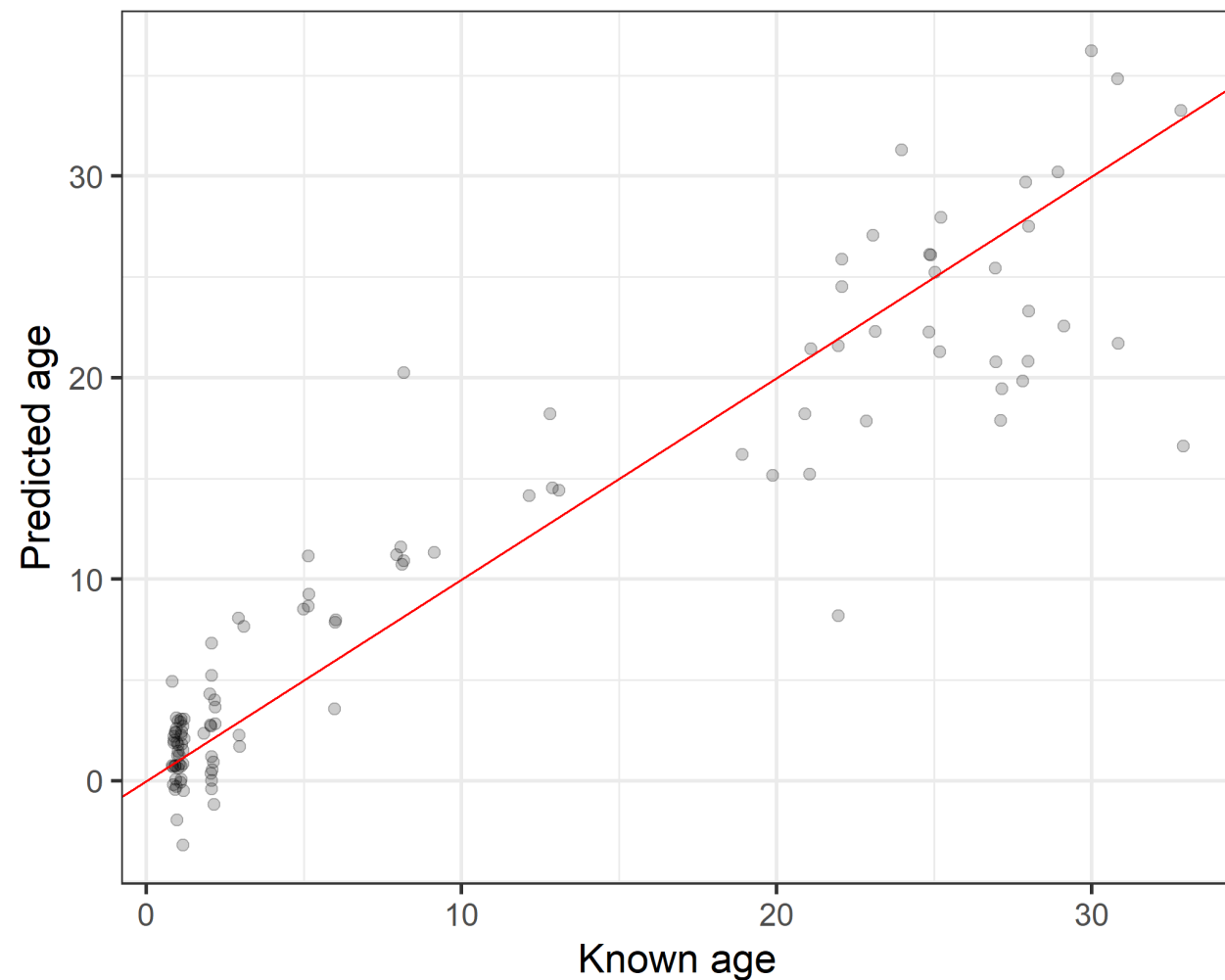
Tooth — Canine — Premolar

Preliminary seal results

Canine



Premolar



Preliminary Summary – Harp seals

- We were able to fit calibration models for harp seal
- As a proof of concept, definitely successful, but we are distant from implementation
- Accuracy and precision are poor vs. fish
 - RMSE and average % CV bias are much higher
- There are some key differences between marine mammals and fish that may require our rethinking how to build calibration models for aging when used for marine mammals
 - Sample sizes may be more limiting
 - Hundreds to thousands of otoliths often available
 - There may only be tens to hundreds of teeth available
 - Given differences in life spans, and aging methods ages for marine mammals are a bit more categorical than continuous integers for fish
 - It may be better to use a classification model designed for sequential categories to age marine mammals (and perhaps long-lived fish) than PLSR
- Further analyses will be performed, this work is very preliminary

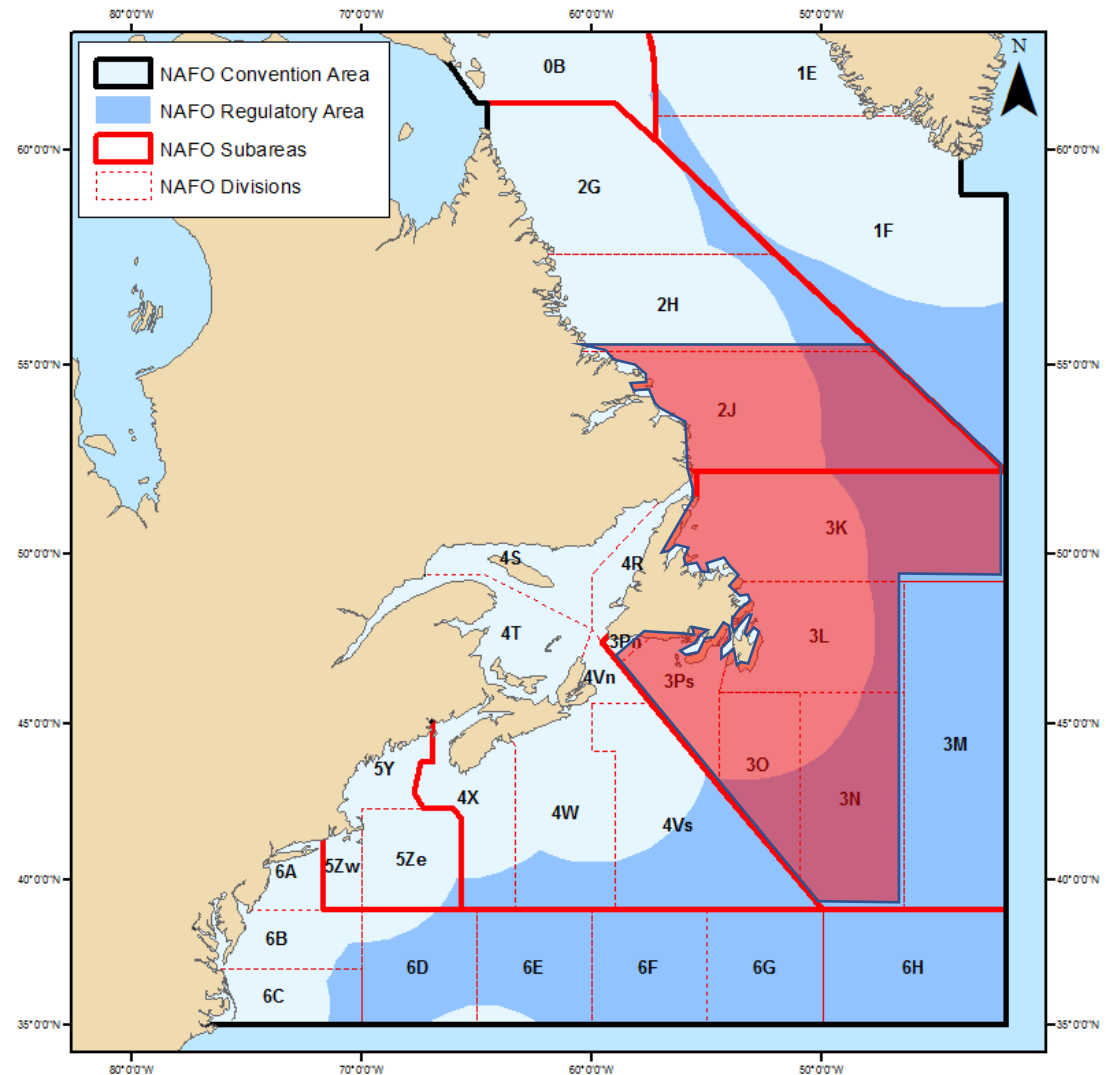
Identifying fish species from otoliths

Introduction

- Benson et al. (2020) showed that FT-NIRs may have fishery applications beyond aging otoliths and other hard parts
- FT-NIRs spectra of fish otoliths can also be used in combination with classification tools to identify the species of fish
- This has a few potential applications
 - Distinguishing between species with very similar appearances
 - **Identifying prey species from predator stomachs**
 - Determining stock ID of fish from commercial samples
- Particularly interested in distinguishing between Atlantic and Greenland cod to better understand harp seal predation

Methods

- Otoliths were collected from 10 species of fish
 - Bulk of samples were **Atlantic cod** (n = 802) and **herring** (603) from an aging study
 - Remainder were **Arctic cod** (99), **capelin** (56), **Greenland cod** (46), **pollock** (88), **Acadian redfish** (70), **Greenland turbot** (70), **witch flounder** (74), and **yellowtail flounder** (70)
- An FT-NIR spectra was collected for each individual using a Bruker Tango FT-NIR spectrometer



Methods

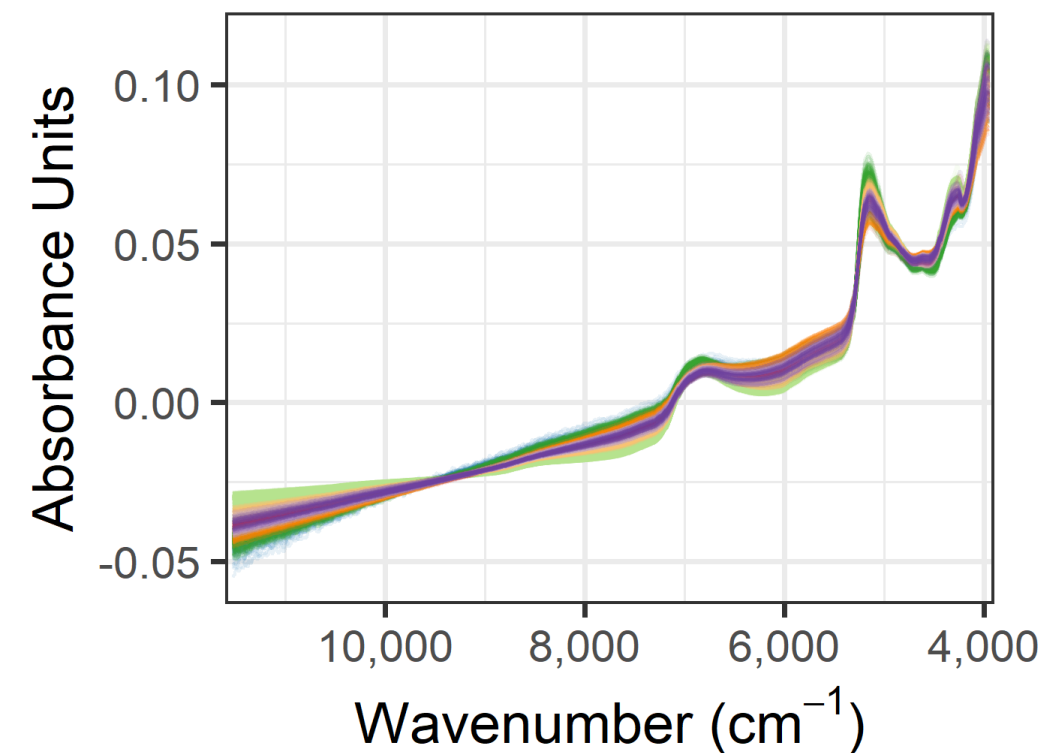
- We then performed QA/QC on the spectra
 - Qualitative examination of PCA plots; Examining Hotellings T2 and Q-residuals
- The QA/QC'd data were then divided into fit (70%) and test (30%) data sets
- We used 4 classification models:
 - Soft-independent modeling of class analogies (SIMCA)
 - Partial least squares discriminant analysis (PLS-DA)
 - K nearest neighbours (KNN)
 - Support Vector Machines (SVM)

Methods

- Metrics used were
 - **Precision/Specificity** – $\text{True Positives} / (\text{True Positives} + \text{False Positives})$
 - **Accuracy** – $(\text{True Positives} + \text{True Negatives}) / \text{All instances}$
 - **Misclassified** – $\text{False Positives} / (\text{True Negatives} + \text{False Positives})$
- The potential for classification models to complete 3 tasks was investigated:
 1. Correctly identify the species an otolith came from given the 10 species (multiclass model – 10 species; testing with Accuracy)
 2. Correctly determine if the otolith came from a Greenland cod (binary model – Greenland cod or other; testing with P/S & Misclassified)
 3. Correctly determine if the otolith came from an Atlantic cod (binary model – Atlantic cod or other ; testing with P/S & Misclassified)

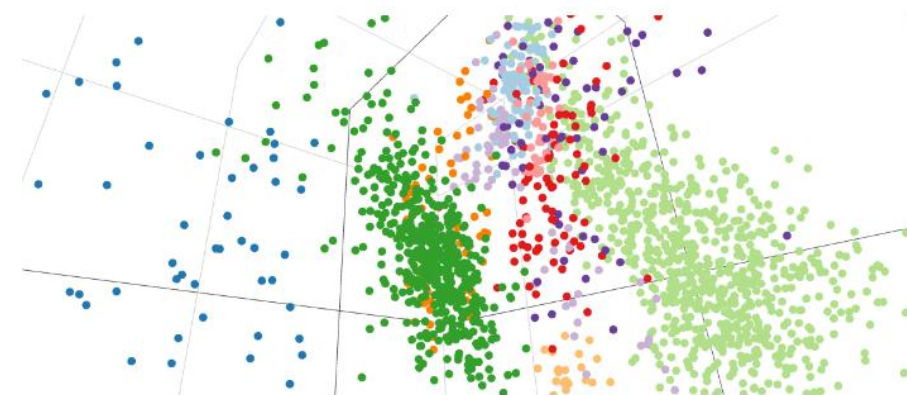
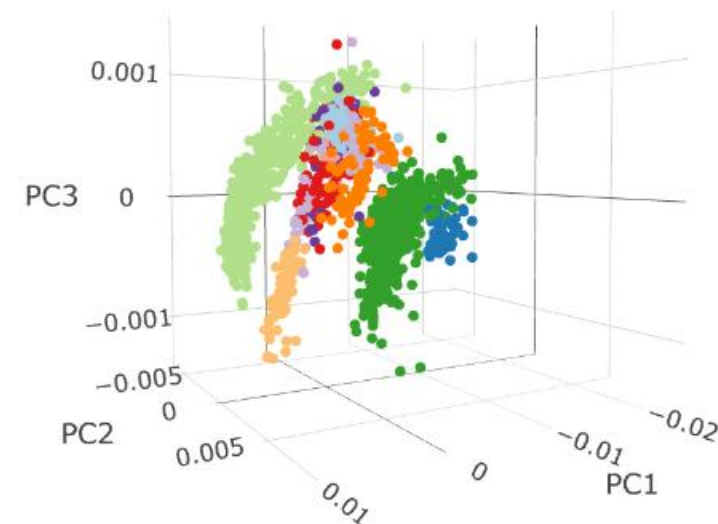
Results – FT-NIR to ID fish species

Absorbance by species

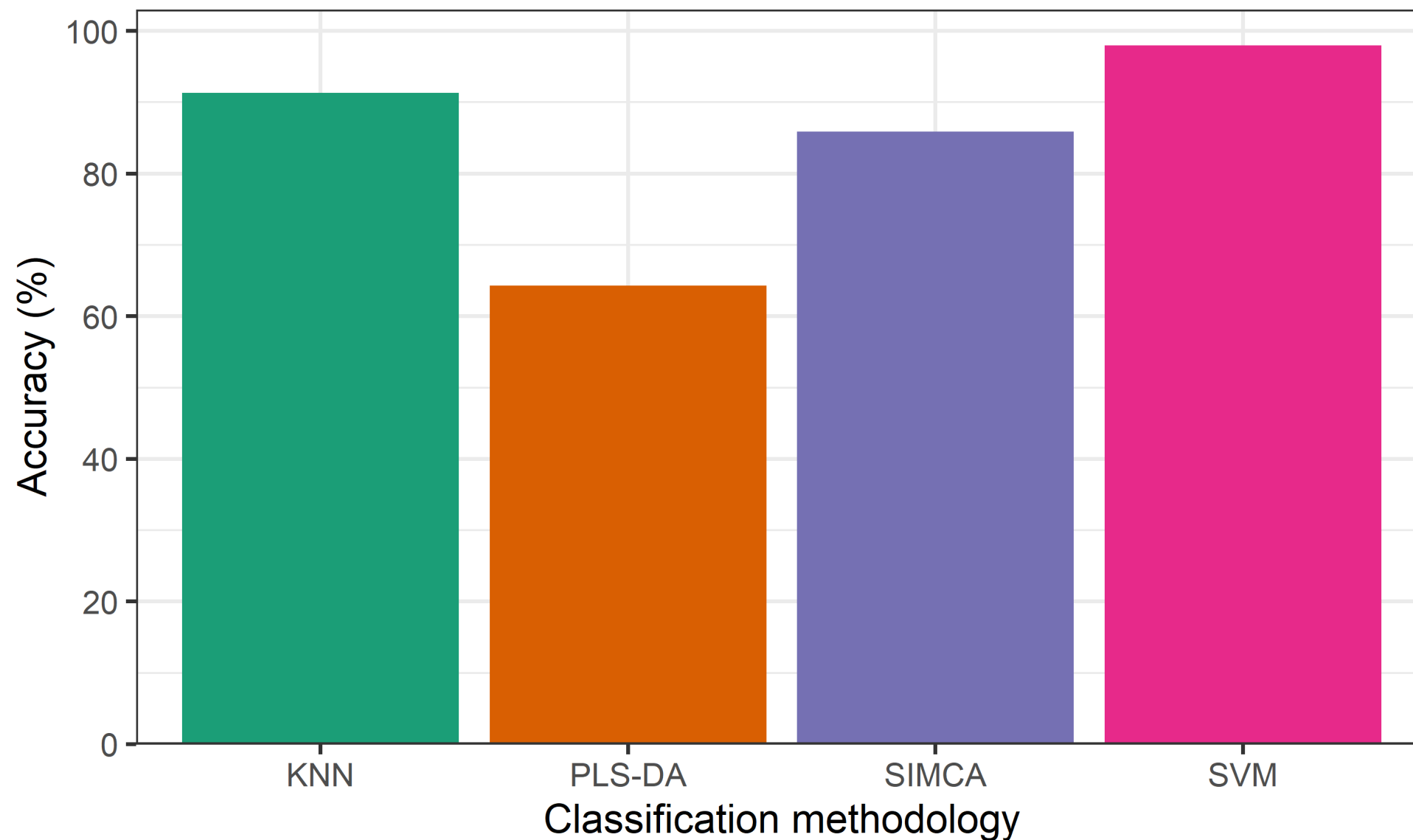


Species

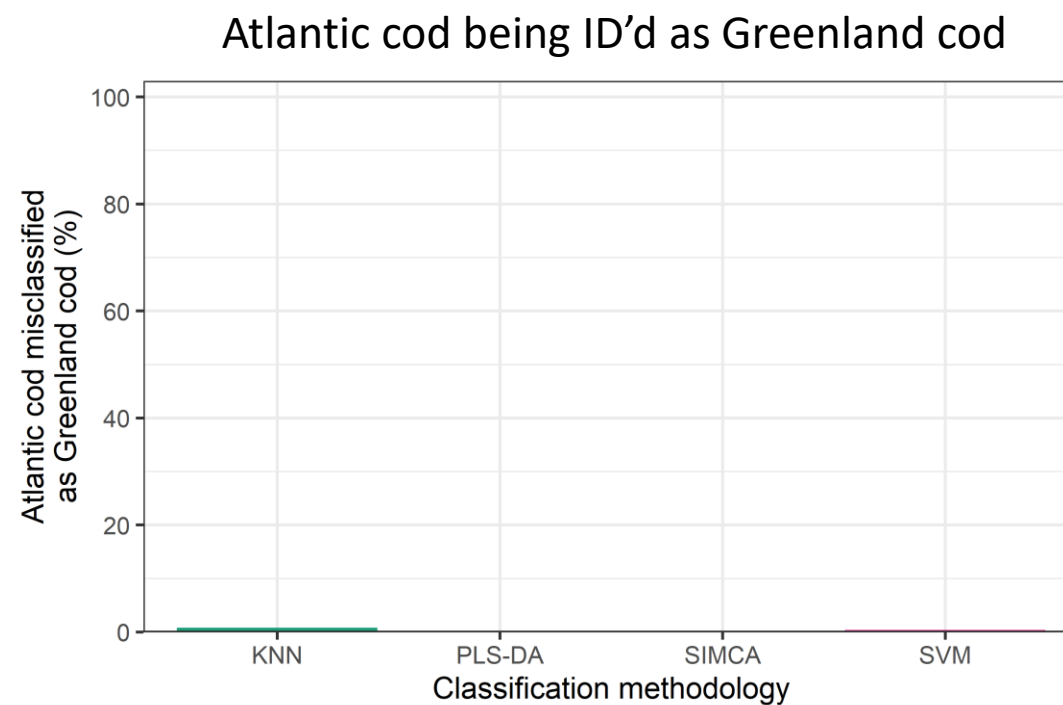
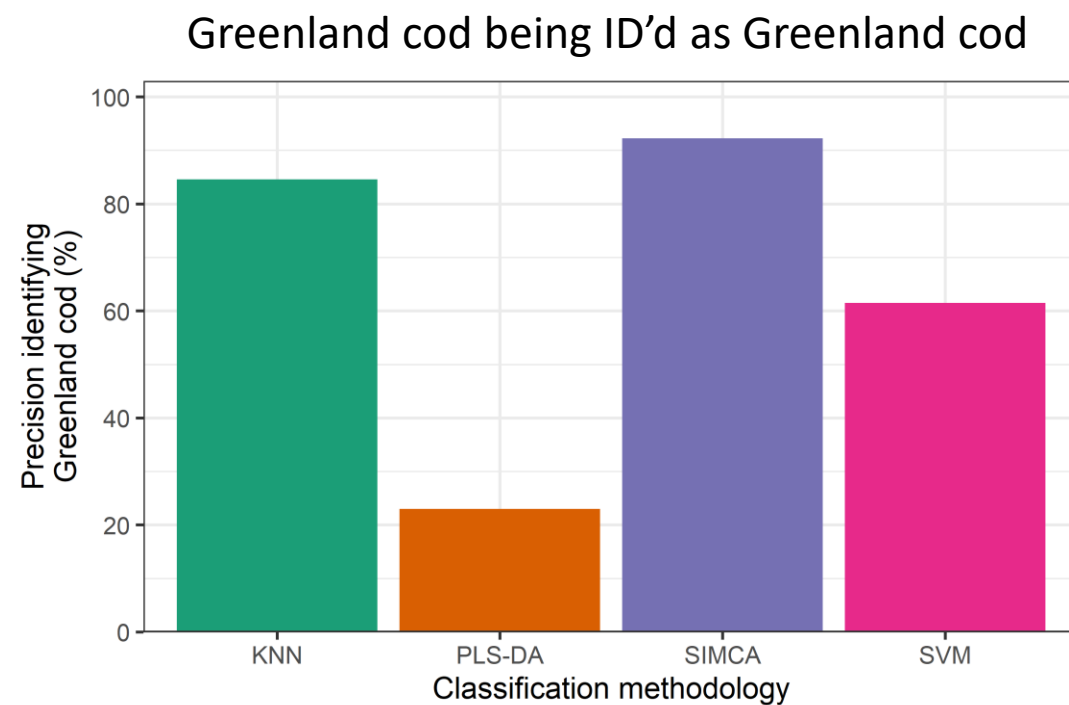
- Arctic cod
- Capelin
- Atlantic cod
- Atlantic herring
- Greenland cod
- Pollock
- Redfish
- Turbot
- Witch flounder
- Yellowtail flounder



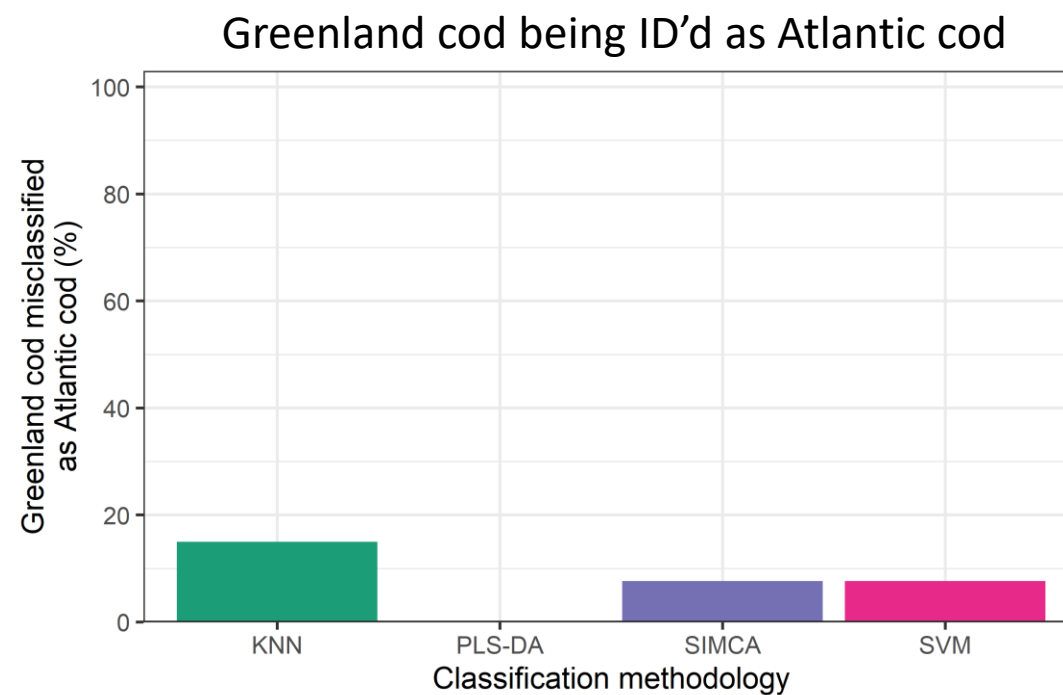
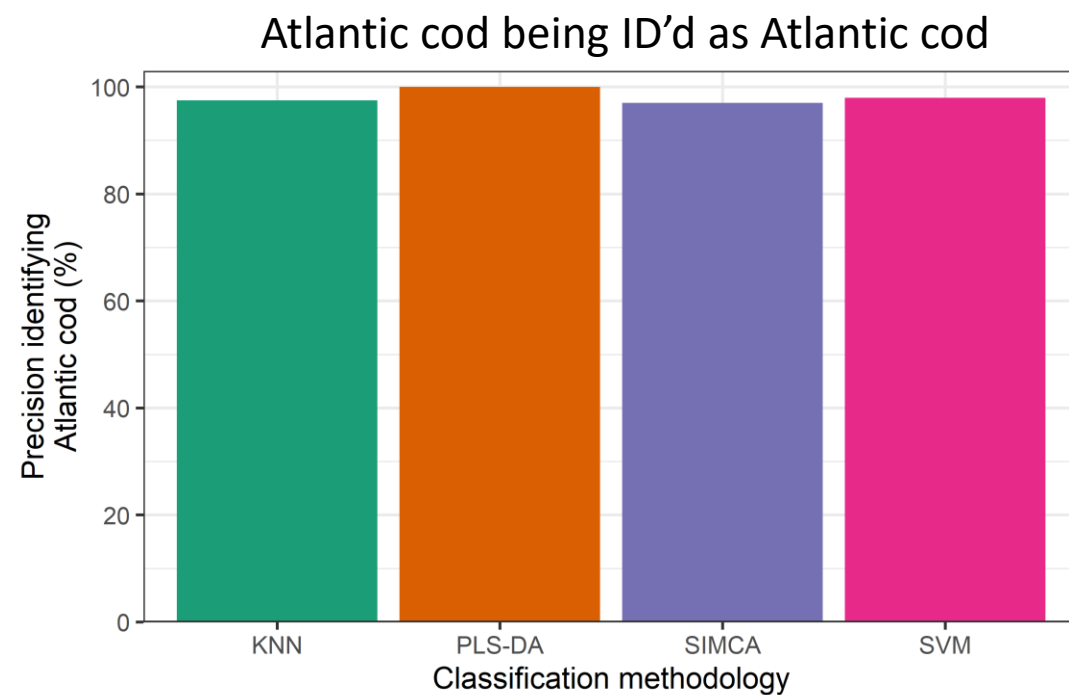
Results – What species is this?



Results – Greenland cod binary class'n model



Results – Atlantic cod binary class'n model



Discussion

- We were successful in developing classification models to identify fish species from otolith FT-NIR spectra (SVM > KNN > SIMCA > PLS-DA)
- Despite a relatively small samples size (~32), we were able to develop models that had high precision for identifying Greenland cod while not misclassifying Atlantic cod using SIMCA
- All models had high precision for identifying Atlantic cod, likely due to large sample size (~561), but did misidentify some Greenland cod as Atlantic cod; PLS-DA performed best (PLS-DA > SIMCA = SVM > KNN)
- Currently investigating whether these methods will work for Atlantic and Greenland cod otoliths that have been artificially digested

On-going work involving NAFC

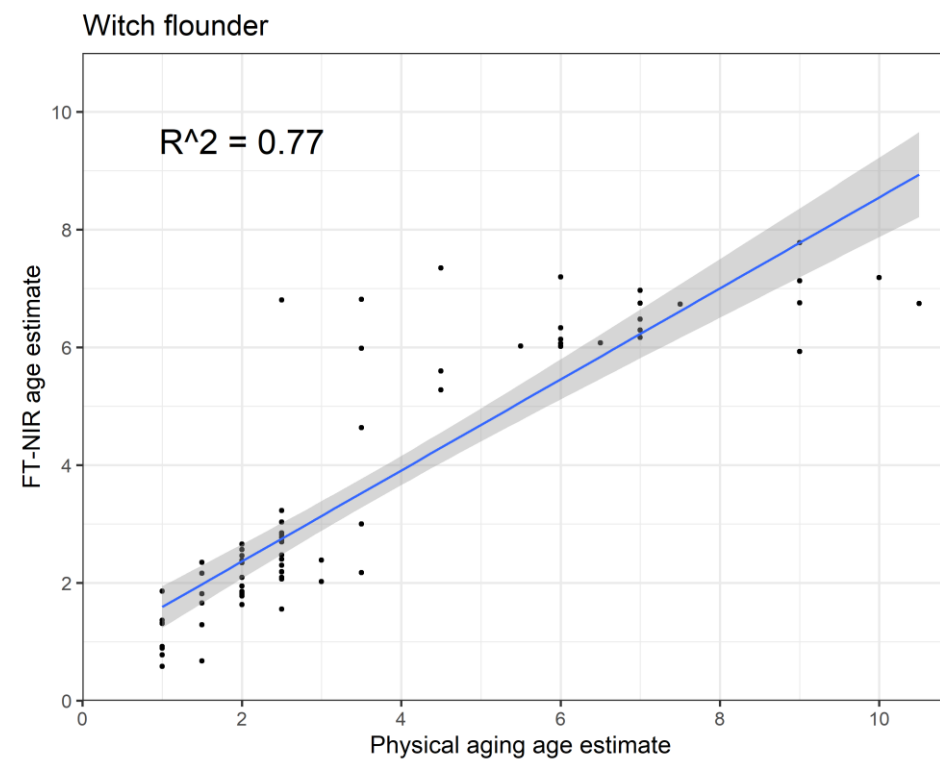
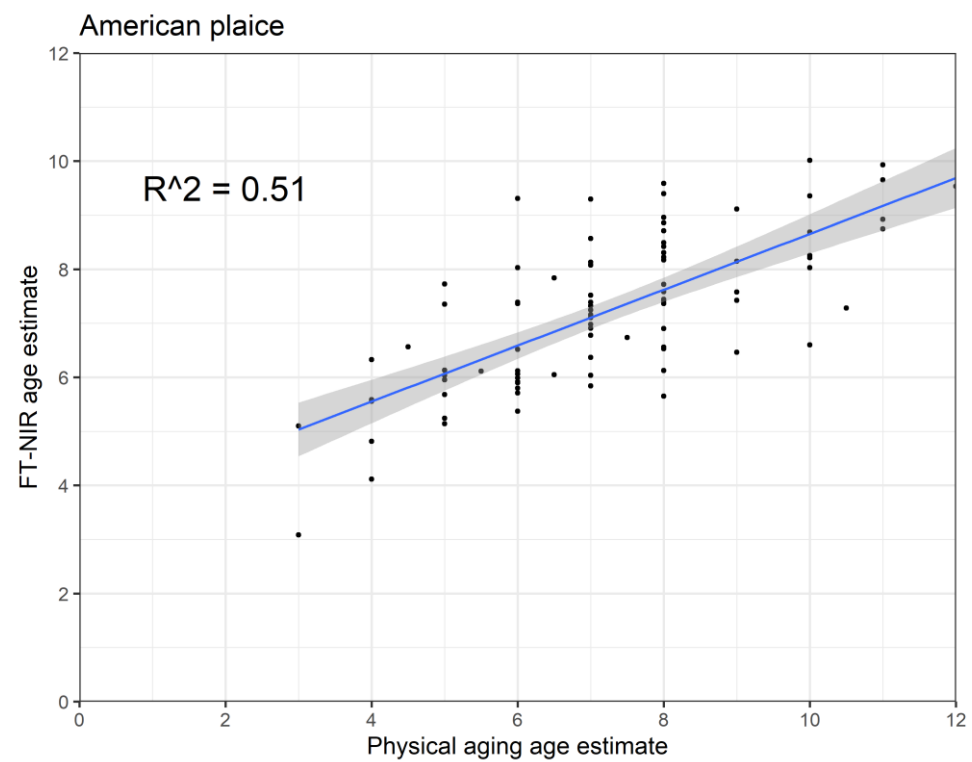
Separating Atlantic from Greenland cod

- Rajeev Kumar is leading a follow-up project to the work I've presented on species ID:
 - Solving hard problems with hard parts: Leveraging otolith spectrometry to enhance species differentiation and age determination of cod in pinniped diets
- This project is focused on juvenile Atlantic and Greenland cod, a preferred food option for harp seals
 - Expanded sampling of Atlantic and Greenland cod otoliths
 - Testing whether we can correctly identify Atlantic and Greenland cod using FT-NIR using both fresh and artificially partially digested otoliths
 - Also testing whether we can use FT-NIR to age partially digested Atlantic cod otoliths

Stock ID in herring

- Karolina Jonko-Sobuś is leading a project involving Canadian and Polish researchers using FT-NIR applied to otoliths to try and determine the stock origin of herring
- Used herring otoliths from the Gulf of St. Lawrence and around NL (NAFO Divs. 3L, 3Ps, 4R, and 4S) and from the Western Baltic Component and from the northern and southern components of Central Baltic herring stocks
- Used both unsupervised and supervised classification models to discriminate herring stocks within the two regions (Western Atlantic and Baltic)
- Manuscript will be submitted soon...

Overall Summary



Overall Summary

- Our work has shown that FT-NIR shows great potential for increasing the speed (and reducing the cost) at which we age Atlantic cod otoliths, with little loss in accuracy or precision
 - Potentially less useful for harp seals
- We're still early in the process of adopting this method for Atlantic cod when looking at the NOAA model e.g. Application Development
- Species ID using FT-NIR is feasible for fresh otoliths, additional work is in progress to test the methodology for partially digested otoliths
- FT-NIR is a new technology for aging, which shows great potential for supporting fish aging programs and investigations into predation